

ANCHORAGE BIRCHWOOD, AK, USA

WMO: 702746

Lat: 61.416N

Lon: 149.507W

Elev: 29

StdP: 100.97

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 26497

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-22.2	-29.7	0.2	-22.8	-26.3	0.3	-20.6	11.0	-4.5	8.3	1.0	0.5	150	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	22.8	15.3	21.3	14.5	19.3	13.5	16.2	21.9	15.1	19.8	14.2	18.3	2.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	9.7	17.9	12.7	9.2	16.9	12.2	8.9	16.4	45.1	21.9	41.9	19.9	39.8	18.4	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	5.3	4.3	-26.6	25.4	2.9	2.3	-28.7	27.1	-30.5	28.5	-32.1	29.8	-34.2	31.5		
(5)			-26.8	17.2	2.6	1.6	-28.7	18.4	-30.2	19.3	-31.6	20.3	-33.5	21.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-9.3	-6.8	-4.1	3.2	9.0	13.4	15.0	13.5	9.1	1.9	-5.7	-8.6	(6)
(7)		DBStd	9.96	7.18	5.76	5.20	3.22	2.86	2.30	1.99	2.13	2.79	4.60	5.85	6.42	(7)
(8)		HDD10.0	3113	599	470	437	206	53	1	0	1	46	253	471	576	(8)
(9)		HDD18.3	5751	858	703	695	455	291	149	107	151	276	511	721	834	(9)
(10)		CDD10.0	408	0	0	0	1	21	102	154	109	20	1	0	0	(10)
(11)		CDD18.3	4	0	0	0	0	0	1	2	1	0	0	0	0	(11)
(12)		CDH23.3	41	0	0	0	0	1	13	22	6	0	0	0	0	(12)
(13)		CDH26.7	5	0	0	0	0	0	1	4	0	0	0	0	0	(13)
(14)	Wind	WSAvg	1.4	1.3	1.3	1.6	1.5	1.9	1.6	1.3	1.2	1.2	1.4	1.4	1.5	(14)
(15)	Precipitation	PrecAvg	457	27	26	21	13	18	28	48	70	85	48	35	38	(15)
(16)		PrecMax	591	55	60	51	46	42	66	92	170	176	118	76	69	(16)
(17)		PrecMin	339	5	3	3	3	5	8	15	28	25	17	2	3	(17)
(18)		PrecStd	62	15	15	11	9	9	14	21	36	41	22	19	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	7.1	7.8	16.1	22.4	25.2	26.2	24.1	18.0	12.9	8.7	7.6	(19)
(20)			MCWB	3.1	3.1	3.7	8.6	12.7	16.0	17.1	16.4	11.7	8.4	4.8	3.8	(20)
(21)		2%	DB	4.0	3.9	5.8	12.4	19.0	22.4	22.8	22.3	16.4	10.9	5.3	4.0	(21)
(22)			MCWB	1.7	1.6	2.3	6.2	11.2	14.4	15.8	15.5	11.0	7.3	3.0	1.9	(22)
(23)		5%	DB	2.3	2.4	3.9	10.9	17.0	20.3	21.2	20.0	15.0	8.9	2.9	2.0	(23)
(24)			MCWB	0.9	0.9	1.0	5.2	9.9	13.6	15.0	14.3	10.6	6.4	1.4	0.7	(24)
(25)		10%	DB	0.4	1.1	2.7	8.8	14.7	18.3	19.1	17.9	13.6	7.7	2.0	-0.3	(25)
(26)			MCWB	-0.6	-0.2	0.3	4.4	8.8	12.4	13.9	13.0	10.0	5.5	0.9	-0.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.6	4.4	9.0	13.4	16.8	18.0	17.3	13.1	9.8	5.0	4.1	(27)
(28)			MCD B	5.7	6.0	6.9	15.4	21.0	23.3	24.7	22.7	16.0	11.9	8.1	7.6	(28)
(29)		2%	WB	2.3	1.9	2.8	6.7	11.5	15.1	16.3	15.9	12.2	7.9	3.4	2.1	(29)
(30)			MCD B	4.1	3.5	4.9	11.2	17.9	21.0	22.1	21.1	14.7	9.4	5.1	4.1	(30)
(31)		5%	WB	0.7	0.9	1.6	5.5	10.4	13.9	15.3	14.7	11.5	6.9	1.9	0.4	(31)
(32)			MCD B	2.1	2.5	3.6	9.6	15.9	19.2	19.8	18.9	13.9	8.4	2.9	1.5	(32)
(33)		10%	WB	-0.8	-0.5	0.5	4.6	9.4	12.9	14.5	13.7	10.7	5.8	0.5	-1.1	(33)
(34)			MCD B	0.2	0.7	2.5	8.5	14.2	17.5	18.5	17.3	13.1	7.6	1.5	-0.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.5	7.7	9.3	9.2	9.0	8.1	7.3	7.8	7.6	6.7	6.2	6.5	(35)
(36)			MCD BR	7.2	7.5	8.3	11.7	12.6	11.9	11.2	11.7	9.6	7.4	6.7	8.4	(36)
(37)			MCWBR	5.5	5.7	5.8	7.0	7.3	6.2	6.1	6.3	5.7	4.8	5.1	6.4	(37)
(38)		5% WB	MCD BR	6.6	7.3	7.9	10.6	11.1	10.2	9.6	10.0	8.0	6.2	6.5	7.8	(38)
(39)			MCWBR	5.2	5.7	5.6	6.5	6.5	5.6	5.4	5.8	4.9	4.3	5.1	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.233	0.286	0.306	0.343	0.364	0.362	0.371	0.353	0.314	0.296	0.243	0.383	(40)	
(41)		taud	2.114	2.222	2.259	2.249	2.288	2.380	2.394	2.453	2.546	2.425	2.099	2.659	(41)	
(42)		Ebn at Noon	493	688	810	845	855	864	845	833	813	686	455	297	(42)	
(43)		Edh at Noon	36	65	89	110	115	107	104	90	68	54	36	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.28	0.98	2.59	4.32	5.18	5.34	4.78	3.79	2.45	1.13	0.41	0.15	(44)	
(45)		RadStd	0.04	0.12	0.25	0.36	0.41	0.33	0.36	0.38	0.31	0.13	0.06	0.02	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		N/A	+1.98	+1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page